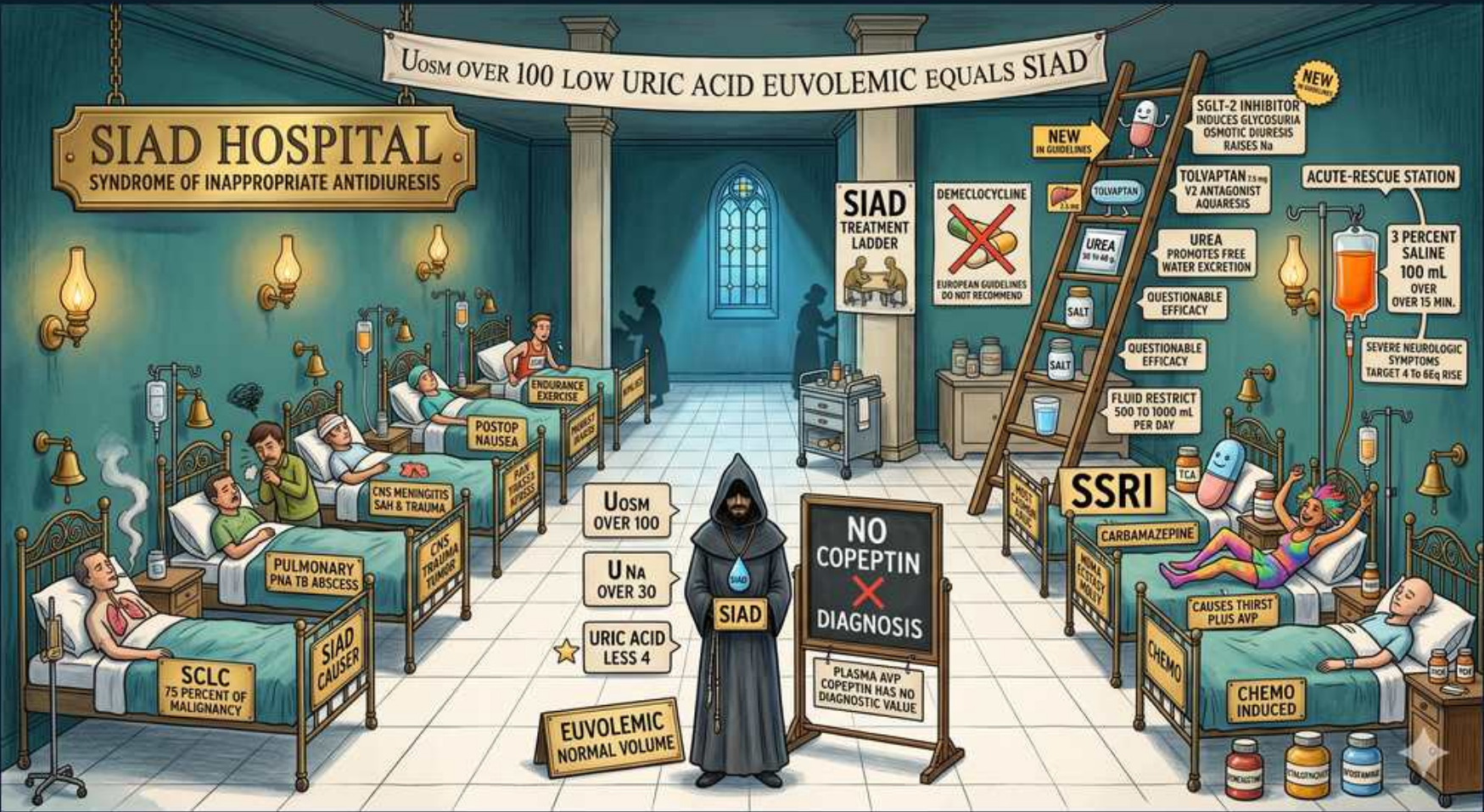


SIADH — euvolemic hyponatremia diagnosis and treatment



1. Euvolemic hypotonic hyponatremia

WHAT YOU SEE

Banner: euvolemic normal volume, SIAD

WHAT IT ENCODES

SIADH is the classic cause of euvolemic hypotonic hyponatremia from inappropriate AVP secretion. Clinically the patient is neither volume depleted nor edematous. Diagnosis requires low serum osmolality with inappropriately concentrated urine.

BOARD TRAP

SIADH requires EUVOLEMIA — hypovolemic (edema-free, dry) or hypervolemic (edematous) hyponatremia are excluded first by exam.

2. Uosm > 100

WHAT YOU SEE

Uosm over 100 criterion

WHAT IT ENCODES

In hypotonic hyponatremia the kidney should maximally dilute urine to <100 mosm/kg. A urine osmolality >100 indicates AVP action and supports SIADH. This reflects failure to excrete free water.

BOARD TRAP

Urine osmolality <100 points to primary polydipsia or low solute intake (beer potomania), NOT SIADH.

3. Urine Na > 30

WHAT YOU SEE

U Na over 30 criterion

WHAT IT ENCODES

Urine sodium >30 mEq/L in a euvolemic patient supports SIADH, indicating the kidney is not avidly retaining sodium. Low urine sodium (<20) suggests hypovolemia or low effective circulating volume. Helps separate SIADH from true volume depletion.

BOARD TRAP

Low urine Na (<20) suggests hypovolemia/hypervolemia with avid Na retention — argues against SIADH despite hyponatremia.

4. Low uric acid (<4)

WHAT YOU SEE

Uric acid less 4 criterion

WHAT IT ENCODES

SIADH causes hypouricemia (uric acid often <4 mg/dL) from dilution and increased urate clearance. Hyperuricemia instead suggests volume depletion. A low uric acid supports the diagnosis.

BOARD TRAP

Hyperuricemia points to volume-depletion hyponatremia; hypouricemia (FEurate high) supports SIADH or cerebral salt wasting.

5. No copeptin diagnostic value

WHAT YOU SEE

No copeptin in diagnosis, plasma AVP/copeptin no diagnostic value

WHAT IT ENCODES

Unlike in DI, measuring AVP/copeptin does not establish SIADH because levels overlap and the diagnosis is clinical/biochemical. SIADH remains a diagnosis of exclusion after ruling out hypothyroidism and adrenal insufficiency. Volume status and the lab criteria define it.

BOARD TRAP

You cannot diagnose SIADH by a high AVP level — always exclude hypothyroidism and cortisol deficiency, which mimic SIADH.

6. SCLC / malignancy cause

WHAT YOU SEE

SCLC lung cancer bed, malignancy

WHAT IT ENCODES

Small cell lung cancer is the classic ectopic AVP source causing paraneoplastic SIADH. Other malignancies (head and neck, duodenal) also produce AVP. New SIADH in a smoker warrants malignancy screening.

BOARD TRAP

Paraneoplastic SIADH in a smoker should trigger a chest CT for small cell lung cancer before labeling idiopathic.

7. CNS / pulmonary causes

WHAT YOU SEE

CNS meningitis SAH trauma and pulmonary pneumonia TB abscess beds

WHAT IT ENCODES

CNS disease (meningitis, subarachnoid hemorrhage, trauma, stroke) and pulmonary disease (pneumonia, TB, abscess) are major non-malignant SIADH causes. They stimulate central AVP release or ectopic production. Often resolve as the underlying illness treats.

BOARD TRAP

In SAH distinguish SIADH from cerebral salt wasting — CSW is hypovolemic with high urine Na, treated with salt/volume not restriction.

8. Drug causes — SSRI, carbamazepine

WHAT YOU SEE

SSRI and carbamazepine drug panel, chemo induced

WHAT IT ENCODES

SSRIs, carbamazepine/oxcarbazepine, cyclophosphamide, and antipsychotics commonly cause drug-induced SIADH, especially in the elderly. Carbamazepine potentiates renal AVP effect. Stopping the drug usually corrects it.

BOARD TRAP

Elderly patient on an SSRI with hyponatremia: drug-induced SIADH is likely — review the med list before extensive workup.

9. Fluid restriction (first-line)

WHAT YOU SEE

Fluid restrict 500 to 1000 mL day

WHAT IT ENCODES

Fluid restriction (often 500-1000 mL/day) is first-line for chronic asymptomatic SIADH. Success depends on a favorable urine-to-plasma electrolyte ratio. It corrects sodium slowly and safely.

BOARD TRAP

Fluid restriction fails when urine osmolality is very high (>500) or urine Na+K exceeds plasma Na — then add salt/urea or a vaptan.

10. Demeclocycline (avoid)

WHAT YOU SEE

Demeclocycline crossed out, questionable efficacy

WHAT IT ENCODES

Demeclocycline induces nephrogenic DI to antagonize AVP but has slow, unpredictable effect and nephrotoxicity. It is now rarely used given safer alternatives. Largely historical.

BOARD TRAP

Demeclocycline is nephrotoxic and slow — not first-line; vaptans or urea are preferred for resistant chronic SIADH.

11. Urea therapy

WHAT YOU SEE

Urea promotes free water excretion panel

WHAT IT ENCODES

Oral urea induces an osmotic diuresis that promotes free water excretion and raises sodium. It is effective and inexpensive for chronic SIADH but limited by palatability. Useful when fluid restriction is insufficient.

BOARD TRAP

Urea is a safe chronic SIADH option (osmotic free-water loss) and carries less overcorrection risk than vaptans.

12. Vaptans (tolvaptan)

WHAT YOU SEE

Tolvaptan V2 antagonist aquaresis, questionable efficacy/cost

WHAT IT ENCODES

Vaptans are V2 receptor antagonists causing selective aquaresis to raise sodium. Tolvaptan carries hepatotoxicity risk and must be started in hospital with sodium monitoring. Cost and overcorrection risk limit use.

BOARD TRAP

Tolvaptan can overcorrect sodium too fast — start inpatient, avoid fluid restriction simultaneously, and watch for osmotic demyelination.

13. 3% hypertonic saline (severe)

WHAT YOU SEE

Acute rescue station, 3 percent saline 100 mL bolus for severe symptoms

WHAT IT ENCODES

Symptomatic severe hyponatremia (seizures, coma) is treated with a 100 mL bolus of 3% saline, repeatable, targeting a 4-6 mEq/L rise. Aim to limit correction to <8-10 mEq/L in 24 hours. Reserved for neurologic emergencies.

BOARD TRAP

Overcorrecting chronic hyponatremia (>8-10 mEq/L/24h) risks osmotic demyelination syndrome — cap the rise and recheck frequently.
